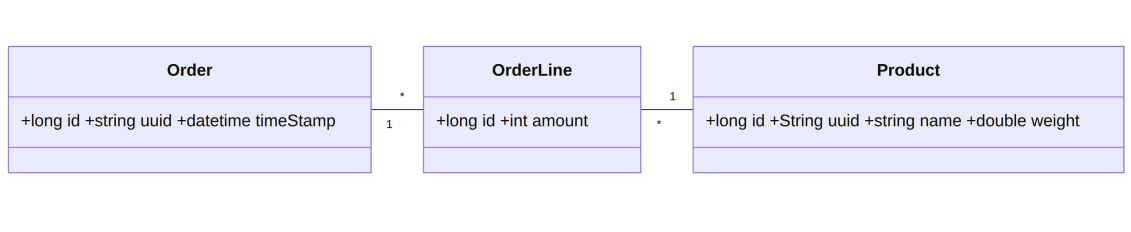
Communication styles

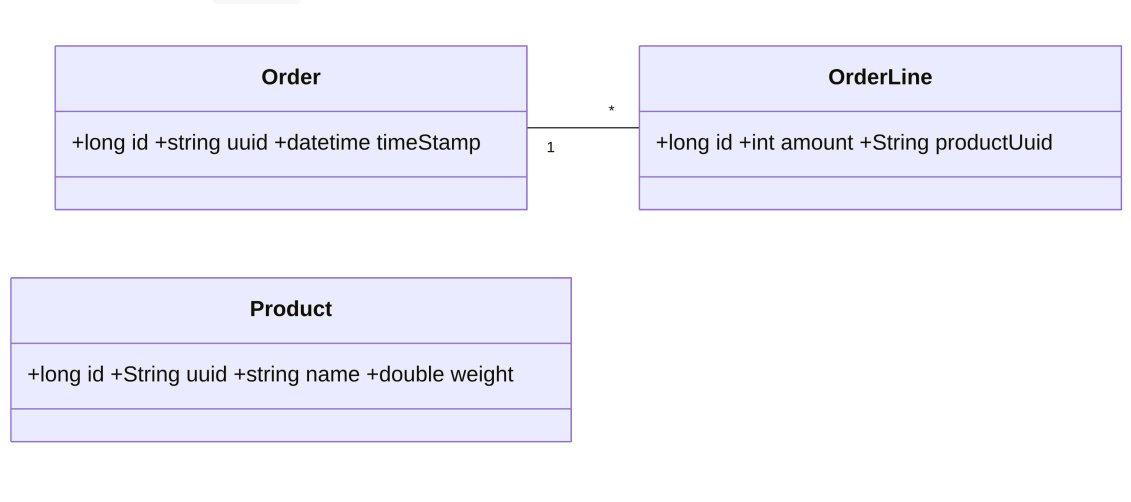
Consequences of a Distributed Data Model

Monolithic architectures store **all data in a single database**, e.g., `Order`, `OrderLine`, and `Product`. Relationships are straightforward:


Order
+long id +string uuid +datetime timeStamp
1
*
OrderLine
+long id +int amount
1
*
Product
+long id +String uuid +string name +double weight

Microservices split data:

- **Order Service:** `Order` + `OrderLine`, without product details.
- **Product Service:** `Product` data separately.


Order
+long id +string uuid +datetime timeStamp
1
*
OrderLine
+long id +int amount +String productUuid
Product
+long id +String uuid +string name +double weight

As a consequence, services must communicate!

Fallacies of Distributed Computing

Designing distributed systems often leads developers to make intuitive assumptions based on single-machine experience. However, **distribution introduces subtle, critical complexities**. The **fallacies of distributed computing** are common misconceptions that seem obvious but fail in real networks. Understanding these fallacies is key to building **robust, scalable, and resilient distributed systems**.

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1. The Network is Reliable

Reality: Networks fail due to hardware, misconfigurations, or transient errors.

```java
@RestController
public class NetworkReliabilityController {

    @PostMapping("/reliable")
    public ResponseEntity<String> reliableService(@RequestBody String data) {
        NetworkService remoteService = new NetworkService();
        String response = remoteService.process(data); // could timeout
        return ResponseEntity.ok(response);
    }
}
```

Mitigation strategies:

- **Retry:** Ensure repeated requests do not corrupt state (idempotent calls).
- **Circuit breaker:** Stop calling failing services temporarily to preserve stability (Hystrix, Resilience4j).

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2. Latency is Zero

Reality: Network calls add significant delays;

- **Latency:** Time for a single request to travel from sender to receiver, including propagation, transmission, and queuing.
- **Round-Trip Time:** Time for a request to go to the receiver and back, including **serialization, processing, network delays, and deserialization**.

Operation	Duration	Normalized
1 CPU cycle	0.3ns	1s
L1 cache access	1ns	3s
L2 cache access	3ns	9s
L3 cache access	13ns	43s
DRAM access	120ns	6min
SSD I/O	0.1ms	4days
HDD I/O	1–10ms	1–12 months
Internet SF → NY	40ms	4years
Internet SF → London	80ms	8years
Internet SF → Sydney	130ms	13years
TCP retransmit	1s	100years
Container reboot	4s	400years

Dependency Injection can be problematic here, as it obscures the true behavior of the `NetworkService`.

```java
@RestController
public class NetworkReliabilityController {
    NetworkService remoteService;

    public NetworkReliabilityController(NetworkService remoteService) {
        this.remoteService = remoteService; // <- minutes
    }

    @PostMapping("/reliable")
    public ResponseEntity<String> reliableService(@RequestBody String data) {
        String response = remoteService.process(data); // <- years ??
        return ResponseEntity.ok(response);
    }
}
```

Mitigation strategies:

- **Batching:** Combine multiple small requests.
- **Caching:** Reduce remote calls (Redis, Memcached).
- **Timeouts:** Set realistic deadlines for service calls.

---

3. Bandwidth is Infinite

Reality: Network throughput is limited; serialization and TCP/IP overhead reduce effective bandwidth.

- 1 Gbps → 128 MB/s → after TCP/IP & serialization → ~32 MB/s

```
Esempio: invio del numero 42 via HTTP/TCP/IP

┌────────────────────────────────────────┐
│ Ethernet header (14 B)                 │
└────────────────────────────────────────┘
┌────────────────────────────────────────┐
│ IP header (20 B)                       │
└────────────────────────────────────────┘
┌────────────────────────────────────────┐
│ TCP header (20 B)                      │
└────────────────────────────────────────┘
┌────────────────────────────────────────┐
│ HTTP headers (~100 B)                  │
│ POST /api/number HTTP/1.1              │
│ Host: example.com                      │
│ Content-Type: application/json         │
│ Content-Length: 13                     │
└────────────────────────────────────────┘
┌────────────────────────────────────────┐
│ JSON body (13 B)                       │
│ {"value":42}                           │
└────────────────────────────────────────┘
┌────────────────────────────────────────┐
│ Dato reale (1-2 B)                     │
│ 42                                     │
└────────────────────────────────────────┘

Totale trasmesso sul cavo: ~167 B
Dati utili: 1–2 B → solo ~1% dei byte è reale dato
```

Mitigation strategies:

- **Backpressure:** Slow consumers can overwhelm faster producers.
- **Throttling & rate limiting:** Use token buckets or leaky bucket algorithms.
- **Compression:** Reduce payload size (Protobuf, Avro).

---

4. The Network is Secure

Reality: Network traffic must be protected with **encryption, authentication, and authorization**.

- **TLS/SSL:** Encrypt transport layer.
- **OAuth2/JWT:** Secure API calls.
- **Audit logging:** Track access to sensitive data.
- **Social engineering risk:** Policies and training are as important as tech controls.

---

5. Topology Does Not Change

Reality: Logical service topology can appear fixed, but in real systems, **dynamic scaling, and network partitions** constantly affect service discovery and connectivity.

Example – Callback scenario:

1. **Service A** registers a **callback** with **Service B** to receive event notifications.
2. **Service B** is removed or fails due to topology changes.
3. **Service A** continues to attempt communication, consuming **threads, CPU, and memory** for a service that no longer exists.

> ⚠️ ***Takeaway:*** *Dynamic topology can lead to wasted resources if callbacks, connections, or retries aren't properly managed.*

Mitigation strategies:

- **Service registries:** Tools like **Eureka, or Kubernetes DNS** enable dynamic discovery of services.
- **Retries and idempotency:** Ensure operations don't cause duplicate side effects during transient failures or service moves.

---

6. There is One Administrator

Reality: Rarely is there a single administrator who knows the entire system. Even if someone designed the network from scratch, over time people leave or change roles, and no one retains complete knowledge.

On top of this, developers often make system behavior **highly configurable** to allow runtime adjustments. Every new switch, lever, or toggle increases flexibility—but also reduces the likelihood that anyone fully understands the system.

> ⚠️ ***Takeaway:*** *The combination of staff turnover and extensive configurability creates an environment where unexpected behaviors and misconfigurations are much more likely.*

Mitigation strategies:

- **Configuration management:** Centralized tools like Spring Cloud Config, Consul, or Vault.
- **Infrastructure as code:** Terraform, Ansible, or Kubernetes manifests reduce human error.
- **Observability:** Logging, metrics, and distributed tracing essential to detect misconfigurations.

---

7. Transport Cost is Zero

Reality: Data movement always incurs **latency and CPU costs**, but in **cloud environments** the impact is even higher. Every time data is moved between nodes:

- You pay for **network bandwidth**.
- You incur **CPU usage** for **serialization and deserialization**.
- These CPU costs translate directly into **monetary cost** on cloud platforms.

> ⚠️ ***Takeaway:*** *Minimizing unnecessary data transfers is critical for both **performance** and **cost efficiency** in cloud systems.*

Mitigation strategies:

- **Serialization:** Protobuf and Avro save bandwidth.
- **Message size optimization:** Avoid sending entire objects when only partial data is needed (GraphQL).

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8. The Network is Homogeneous

Reality: Nodes vary in **language, platform, database, and protocol**.

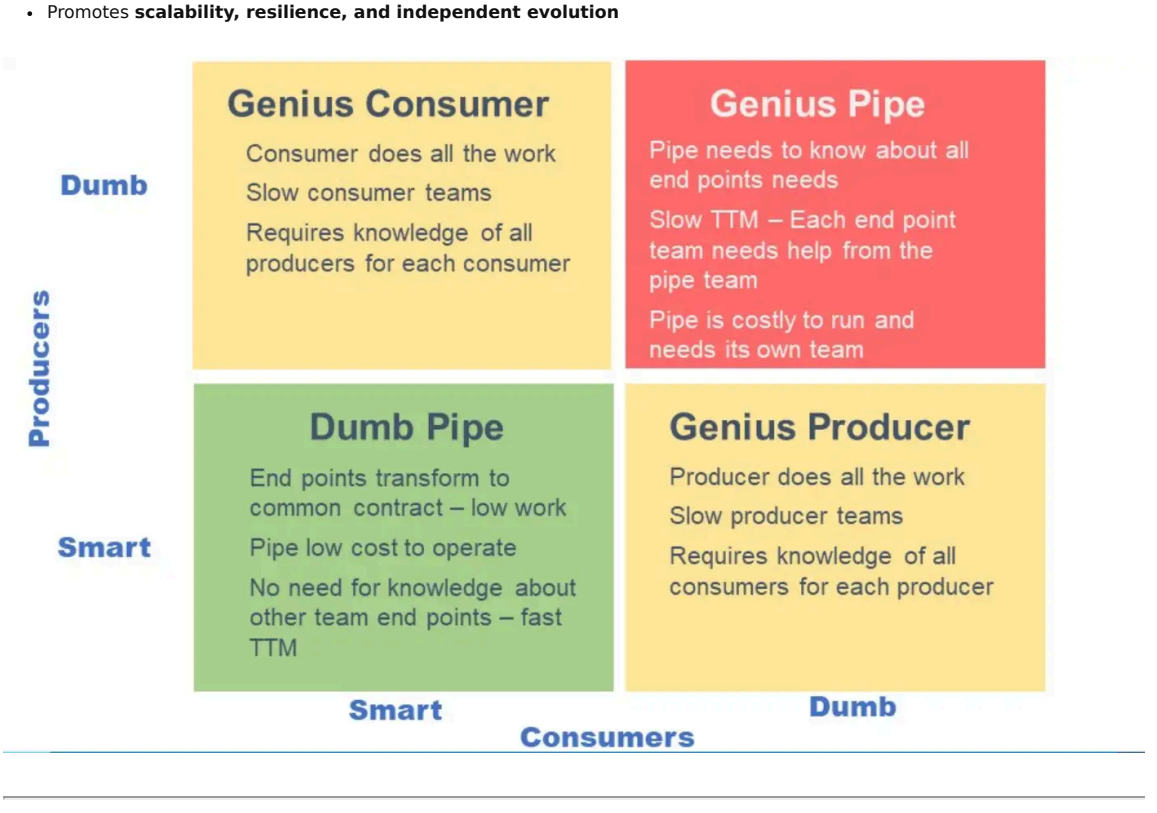
Mitigation strategies:

- **Polyglot persistence:** Mix of relational and NoSQL databases (Postgres, MongoDB, Cassandra).
- **Cross-language communication:** REST, gRPC, or message brokers handle interoperability.
- **Data serialization challenges:** JSON, Protobuf, Avro may have different defaults.

Smart Endpoints, Dumb Pipes

Principle: Keep the **communication layer simple** and put **business logic in services**.

- Avoids complex ESBs and orchestration layers
- Promotes **scalability, resilience, and independent evolution**


Genius Consumer
Consumer does all the work
Slow consumer teams
Requires knowledge of all producers for each consumer
Genius Pipe
Pipe needs to know about all end points needs
Slow TTM – Each end point team needs help from the pipe team
Pipe is costly to run and needs its own team
Dumb
Producers
Smart
Dumb Pipe
End points transform to common contract – low work
Pipe low cost to operate
No need for knowledge about other team end points – fast TTM
Genius Producer
Producer does all the work
Slow producer teams
Requires knowledge of all consumers for each producer
Smart
Dumb
Consumers

Taxonomy of Client-Service Interactions

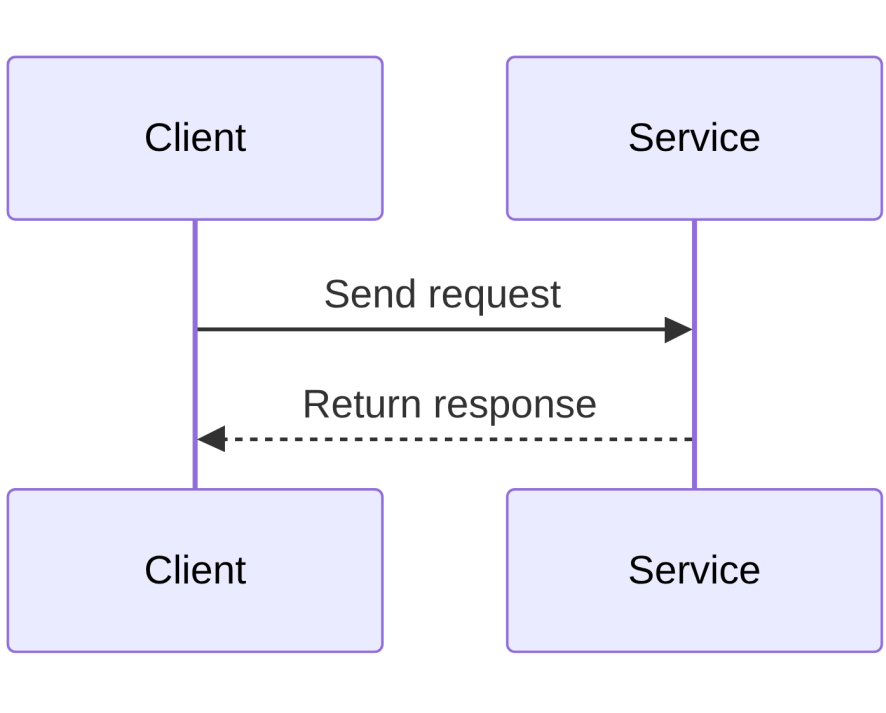
Relationship

- **One-to-One:** Single client interacts with one service; simple but tightly coupled.
- **One-to-Many:** Single client invokes multiple services; allows flexibility and scaling.

Response Timing

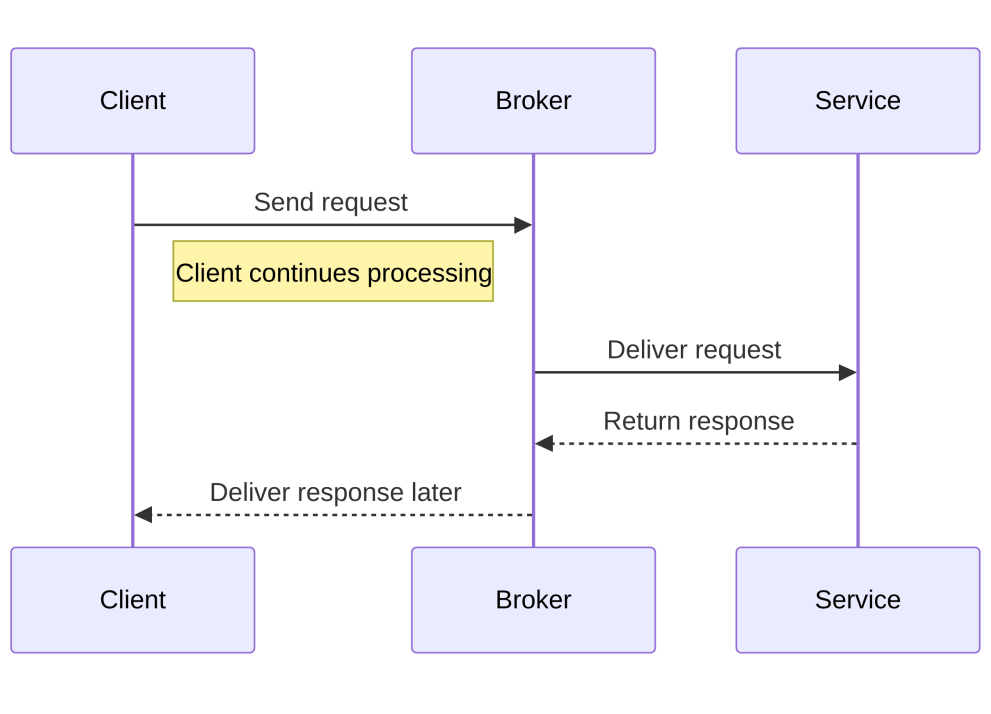
- **Synchronous:** Client blocks; simple but latency accumulates.
- **Asynchronous:** Non-blocking; improves responsiveness but adds complexity.

Request/Response (One-to-One, Sync)


Client
Service
Send request
Return response
Client
Service

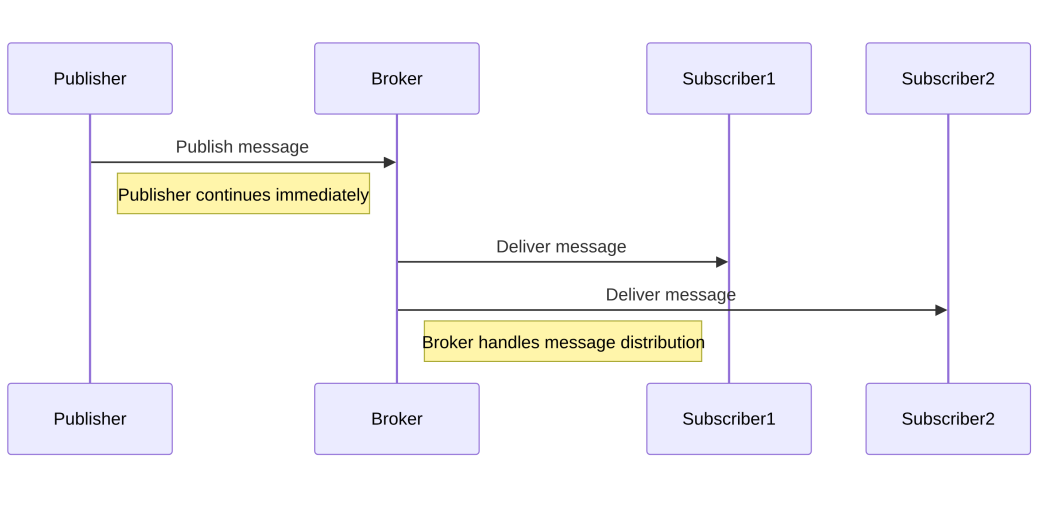
- Classic blocking call.
- Client waits until the service responds.

Async Request/Response (One-to-One, Async)


Client
Broker
Service
Send request
Client continues processing
Deliver request
Return response
Deliver response later
Client
Broker
Service

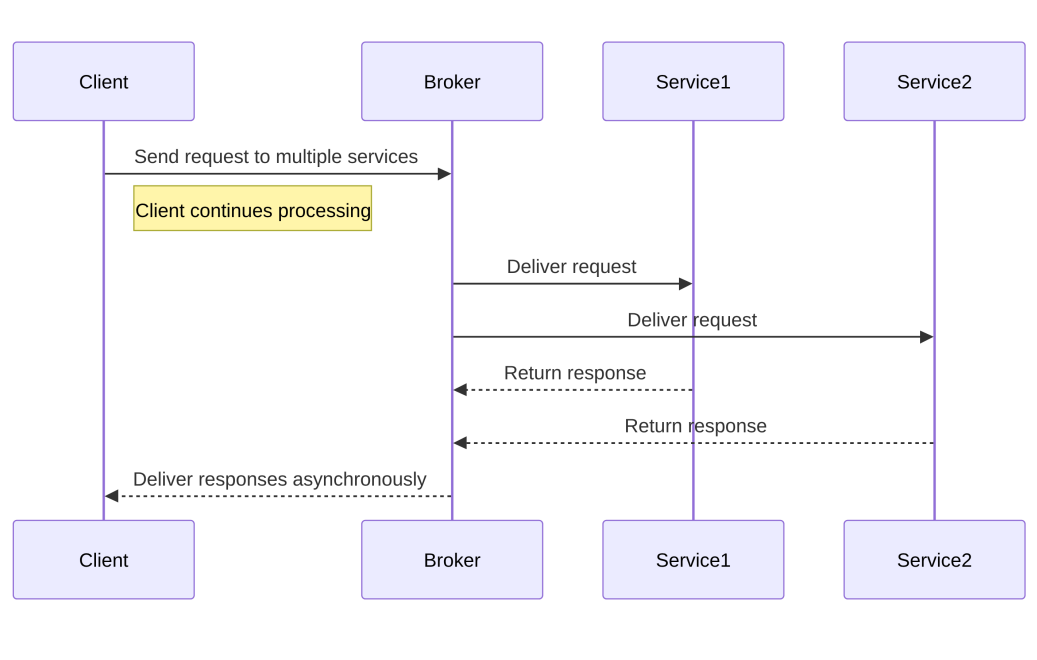
- Non-blocking.
- Client can do other work while waiting for a response.

Publish/Subscribe (One-to-Many, Async)


Publisher
Broker
Subscriber1
Subscriber2
Publish message
Publisher continues immediately
Deliver message
Deliver message
Broker handles message distribution
Publisher
Broker
Subscriber1
Subscriber2

- Publisher sends messages to a broker.
- Multiple subscribers consume messages independently.
- Decouples publisher and consumers.

Publish/Async Responses (One-to-Many, Async)


Client
Broker
Service1
Service2
Send request to multiple services
Client continues processing
Deliver request
Deliver request
Return response
Return response
Deliver responses asynchronously
Client
Broker
Service1
Service2

- Client sends requests to multiple services asynchronously.
- Responses arrive later without blocking the client.

Resources

- *Microservices Patterns*, Chapter 3
- Fallacies of Distributed Systems